

ANDY TRAN

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EDUCATION

Stony Brook University

Bachelor of Science in Computer Science

Stony Brook, NY

Aug 2023 – May 2027

Expected Graduation: May 2027

Relevant Coursework: Operating Systems, Computer Networks, Systems Design, Data Structures and Algorithms, AI, Machine Learning, Databases

TECHNICAL SKILLS

Languages: C, C++, Rust, Python, Java, CUDA, Go, TypeScript, JavaScript, SQL, ARM/MIPS Assembly

Embedded & Automotive: STM32F4 bare-metal firmware, CAN / CAN FD, UDS diagnostics, DBC, DMA, UART/SPI/USB, interrupts & timers, SWD debugging, hardware-in-the-loop testing, panda safety models

Systems & AI/ML: Linux systems programming, real-time scheduling, concurrency, POSIX sockets, PCIe, protocol design, PyTorch, TensorRT, tinygrad, Jetson Orin, camera calibration, sensor fusion

Web & Cloud: React, Next.js, Node.js, AWS (Lambda, DynamoDB, S3), Docker, CI/CD

PROJECTS

End-to-End Autonomous Driving Stack

C, C++, Python, CUDA

github.com/Sontran0118/End-To-End-Self-Driving

2026

- Architected a **9-layer control stack** — camera capture, TensorRT inference, message bus, lateral/longitudinal control, radar handover, USB transport, STM32 firmware, safety enforcement, CAN — that steers and brakes a production car on public roads.
- Built the **lateral loop at 50–70 Hz**: quadratic fit over the model's predicted path validated against known arcs, converted to steering torque on CAN **0x243** with learned offset correction and lane-change logic.
- Designed **longitudinal control** over raw network acceleration — ceiling clamps, jerk limits, and a 5 mph **coast band** before brake engagement — plus a **UDS radar handover** where the factory ECU runs ACC entry and the host assumes authority across a 0.3 s overlap.
- Wrote bare-metal **STM32F407 firmware** as the final gate — transmit allowlists, torque and rate limits, engagement rules, DMA-driven **1.5 Mbaud UART at 2,325 CAN frames/s** — and split model from CAN into separate processes so GPU work can never stall actuation.
- Killed a **99.8% packet-loss bug** wedging the link after 11–61 min: instrumentation isolated a **16-instruction race in USB_WritePacket** enabling an endpoint before its FIFO filled.
- Found **safety code compiled but dead** — unlocked TIM2/TIM9 silently zeroed every heartbeat, timeout, and watchdog check — and proved a hardcoded 1.8 s follow rule **overrode the model in 59% of frames**, braking where it asked to accelerate.

Repo RFQ Engine — Securities Financing

Go, MySQL, Angular

github.com/Sontran0118/repo-rfq-engine

2026

- Guaranteed **exactly-once trade execution under concurrent acceptance** — two dealers can never double-book one quote — via **SELECT ... FOR UPDATE** row locks, 409 state re-validation, and a **UNIQUE** constraint on **trades.rfq_id**.
- Proved it under the Go race detector: **20 concurrent goroutines** at one RFQ yield exactly **1 trade and 19 clean rejections**.
- Eliminated money-precision drift using **int64** minor units and integer basis points (ACT/360 interest, haircut-adjusted collateral), reaching **91% coverage** on an I/O-free domain layer.

Unix File System & PCIe Transaction Layer

C, C++

github.com/Sontran0118/linux-filessystem

2025

- Built a disk-backed Unix file system in C — i-node allocation over a bitmap, direct and single-indirect addressing, full path traversal, and the complete file/directory API — verified by **50+ tests**.
- Parsed **PCIe transaction-layer packets** at the bit level in C++ (byte-enable masking, tag-based tracking of outstanding reads), zero-copy over the caller's buffer: **10K packets in ~10 ms**.

Networked Systems — Protocol Design & Concurrent Servers

C, Sockets

github.com/Sontran0118/AFLENT-Network-Protocol-Implementation

2025

- Designed **AFLENT**, a protocol with a packed **3-byte header** (6-bit array ID, 5-bit fragment, 10-bit length, 3 flags) parsed by bitmasking, reassembling **64 concurrent transfers** correctly from **out-of-order delivery**.
- Wrote a block cipher expanding a 64-bit key to **1024 bits** of round state, and a Texas Hold'em server multiplexing **six players** without blocking over POSIX sockets.

EXPERIENCE

Office Assistant — Roth Quad Area Office

Campus Residences, Stony Brook University

Stony Brook, NY

May 2024 – May 2026

- Resolved first-line IT issues across five residential buildings — network, login, and hardware faults — escalating to the campus DoIT help desk, and maintained office workstations, printers, and peripherals.
- Wrote troubleshooting guides for the most common resident issues, cutting repeat questions and shortening onboarding for new student staff.